

Тема: Настройка виртуальной локальной сети (VLAN)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между

коммутаторами использовать Native VLAN

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

3) Перехватить в WireShark пакеты с тегами и без тегов (nb!), результаты задокументировать

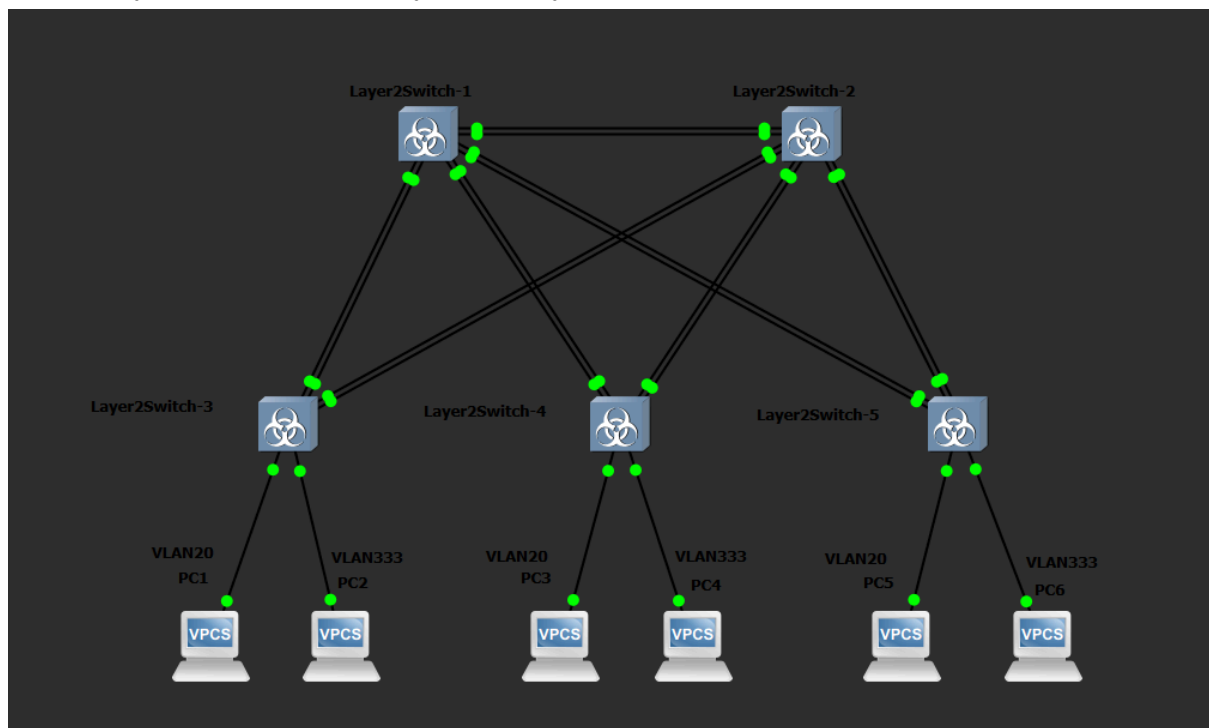
4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

5*) Опциональное задание: Добавить в схему маршрутизатор, подключенный к коммутаторам Layer2Switch1 и Layer2Switch2, настроить через него маршрутизацию между VLAN

Полезная информация: избыточные физические каналы можно поместить в отдельные VLAN и обойтись без STP

1.

Сделал дубликат схемы и запустил все узлы



Необходимо на каждом коммутаторе создать VLAN следующими командами:

```
enable
configure terminal
vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit
end
write memory
```

SW1:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#end
Switch#
*May 26 15:12:41.148: %SYS-5-CONFIG_I: Configured from console by consolewrite m
emory
Switch#write memory
```

SW2:

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#end
Switch#write memory
Building configuration...
Compressed configuration from 5193 bytes to 1975 bytes
*May 26 15:09:45.533: %SYS-5-CONFIG_I: Configured from console by console[OK]
*May 26 15:09:54.043: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
Switch#show vlan brief
*May 26 15:09:55.244: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.

```

VLAN	Name	Status	Ports
1	default	active	Gil/2, Gil/3, Gi2/0
20	VLAN20	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

SW3:

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#end
Switch#
*May 26 15:12:24.346: %SYS-5-CONFIG_I: Configured from console by console
Switch#write memory

```

SW4:

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#end
Switch#
*May 26 15:12:17.501: %SYS-5-CONFIG_I: Configured from console by console
Switch#write memory
Building configuration...

```

SW5:

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 333
Switch(config-vlan)#name VLAN333
Switch(config-vlan)#exit
Switch(config)#end
Switch#write memory
*May 26 15:17:29.060: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...

```

Также были добавлены еще vlan999 чтобы не использовать STP.

enable

configure terminal

vlan 999

name UNUSED_LINKS

exit

end

write memory

Настройка портов свичей:

Layer2Switch-1

Gi0/0 -> Layer2Switch-2 Gi0/0 рабочий trunk

Gi0/1 -> Layer2Switch-2 Gi0/1 лишний trunk VLAN999

Gi0/2 -> Layer2Switch-3 Gi0/0 рабочий trunk

Gi0/3 -> Layer2Switch-3 Gi0/1 лишний trunk VLAN999

Gi1/0 -> Layer2Switch-4 Gi0/0 лишний trunk VLAN999

Gi1/1 -> Layer2Switch-4 Gi0/1 лишний trunk VLAN999

Gi1/2 -> Layer2Switch-5 Gi0/0 лишний trunk VLAN999

Gi1/3 -> Layer2Switch-5 Gi0/1 лишний trunk VLAN999

Настройка интерфейсов:

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#

```
Switch(config)#hostname Layer2Switch-1
Layer2Switch-1(config)#
Layer2Switch-1(config)#vlan 999
Layer2Switch-1(config-vlan)# name UNUSED_LINKS
Layer2Switch-1(config-vlan)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet0/0
Layer2Switch-1(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-2_Gi0/0
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 333
Layer2Switch-1(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet0/2
Layer2Switch-1(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-3_Gi0/0
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 333
Layer2Switch-1(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet0/1
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-2_Gi0/1
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet0/3
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-3_Gi0/1
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet1/0
```

```
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-4_Gi0/0
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet1/1
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-4_Gi0/1
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet1/2
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-5_Gi0/0
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#interface GigabitEthernet1/3
Layer2Switch-1(config-if)# description
UNUSED_LINK_TO_Layer2Switch-5_Gi0/1
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-1(config-if)# switchport mode trunk
Layer2Switch-1(config-if)# switchport trunk native vlan 999
Layer2Switch-1(config-if)# switchport trunk allowed vlan 999
Layer2Switch-1(config-if)# no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
Layer2Switch-1(config)#end
Layer2Switch-1#write memory
*May 26 16:22:30.655: %PLATFORM-5-SIGNATURE_VERIFIED: Image
'flash0:/vios_l2-adventerprisek9-m' passed code signing
verification
```

show vlan brief

show interfaces trunk

```
Layer2Switch-1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi2/0
20	VLAN20	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	
999	UNUSED_LINKS	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

```
Layer2Switch-1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	on	802.1q	trunking	999
Gi0/2	on	802.1q	trunking	333
Gi0/3	on	802.1q	trunking	999
Gi1/0	on	802.1q	trunking	999
Gi1/1	on	802.1q	trunking	999
Gi1/2	on	802.1q	trunking	999
Gi1/3	on	802.1q	trunking	999

```
Port Vlan
```

Port	Vlans allowed on trunk
Gi0/0	20,333
Gi0/1	999
Gi0/2	20,333
Gi0/3	999
Gi1/0	999
Gi1/1	999
Gi1/2	999
Gi1/3	999

```
Port Vlan
```

Port	Vlans allowed and active in management domain
Gi0/0	20,333
Gi0/1	999
Gi0/2	20,333
Gi0/3	999
Gi1/0	999
Gi1/1	999
Gi1/2	999
Gi1/3	999

```
Port Vlan
```

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20,333
Gi0/1	999
Gi0/2	20,333
Gi0/3	999
Gi1/0	999
Gi1/1	none
Gi1/2	999
Gi1/3	999

Layer2Switch-2

Gi0/0 -> Layer2Switch-1 Gi0/0 рабочий trunk
Gi0/1 -> Layer2Switch-1 Gi0/1 лишний trunk VLAN999
Gi0/2 -> Layer2Switch-3 Gi0/2 лишний trunk VLAN999
Gi0/3 -> Layer2Switch-3 Gi0/3 лишний trunk VLAN999
Gi1/0 -> Layer2Switch-4 Gi0/2 рабочий trunk
Gi1/1 -> Layer2Switch-4 Gi0/3 лишний trunk VLAN999
Gi1/2 -> Layer2Switch-5 Gi0/2 рабочий trunk
Gi1/3 -> Layer2Switch-5 Gi0/3 лишний trunk VLAN999

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Layer2Switch-2
Layer2Switch-2(config)#
Layer2Switch-2(config)#vlan 999
Layer2Switch-2(config-vlan)# name UNUSED_LINKS
Layer2Switch-2(config-vlan)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet0/0
Layer2Switch-2(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-1_Gi0/0
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 333
Layer2Switch-2(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet1/0
Layer2Switch-2(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-4_Gi0/2
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 333
Layer2Switch-2(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet1/2
Layer2Switch-2(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-5_Gi0/2
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 333
Layer2Switch-2(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
```

```
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet0/1
Layer2Switch-2(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gi0/1
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 999
Layer2Switch-2(config-if)# switchport trunk allowed vlan 999
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet0/2
Layer2Switch-2(config-if)# description
UNUSED_LINK_TO_Layer2Switch-3_Gi0/2
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 999
Layer2Switch-2(config-if)# switchport trunk allowed vlan 999
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet0/3
Layer2Switch-2(config-if)# description
UNUSED_LINK_TO_Layer2Switch-3_Gi0/3
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 999
Layer2Switch-2(config-if)# switchport trunk allowed vlan 999
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet1/1
Layer2Switch-2(config-if)# description
UNUSED_LINK_TO_Layer2Switch-4_Gi0/3
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 999
Layer2Switch-2(config-if)# switchport trunk allowed vlan 999
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#interface GigabitEthernet1/3
Layer2Switch-2(config-if)# description
UNUSED_LINK_TO_Layer2Switch-5_Gi0/3
Layer2Switch-2(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-2(config-if)# switchport mode trunk
Layer2Switch-2(config-if)# switchport trunk native vlan 999
Layer2Switch-2(config-if)# switchport trunk allowed vlan 999
```

```
Layer2Switch-2(config-if)# no shutdown
Layer2Switch-2(config-if)#exit
Layer2Switch-2(config)#
Layer2Switch-2(config)#end
Layer2Switch-2#
*May 26 16:14:08.543: %SYS-5-CONFIG_I: Configured from console by
console
Layer2Switch-2#wr
Building configuration...
Compressed configuration from 6685 bytes to 2609 bytes[OK]
*May 26 16:14:37.064: %GRUB-5-CONFIG_WRITING: GRUB configuration
is being updated on disk. Please wait...
*May 26 16:14:38.065: %GRUB-5-CONFIG_WRITTEN: GRUB configuration
was written to disk successfully.
```

show vlan brief

show interfaces trunk

Layer2Switch-2#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gi2/0
20	VLAN20	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	
999	UNUSED_LINKS	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Layer2Switch-2#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.lq	trunking	333
Gi0/1	on	802.lq	trunking	999
Gi0/2	on	802.lq	trunking	999
Gi0/3	on	802.lq	trunking	999
Gi1/0	on	802.lq	trunking	333
Gi1/1	on	802.lq	trunking	999
Gi1/2	on	802.lq	trunking	333
Gi1/3	on	802.lq	trunking	999

Port Vlan allowed on trunk

Gi0/0	20,333
Gi0/1	999
Gi0/2	999
Gi0/3	999
Gi1/0	20,333
Gi1/1	999
Gi1/2	20,333
Gi1/3	999

Port Vlan allowed and active in management domain

Gi0/0	20,333
Gi0/1	999
Gi0/2	999
Gi0/3	999
Gi1/0	20,333
Gi1/1	999
Gi1/2	20,333
Gi1/3	999

Port Vlan in spanning tree forwarding state and not pruned

Gi0/0	20,333
Gi0/1	none
Gi0/2	999
Gi0/3	999
Gi1/0	20,333
Gi1/1	999
Gi1/2	20,333
Gi1/3	999

Layer2Switch-3

Gi0/0 -> Layer2Switch-1 Gi0/2 рабочий trunk

Gi0/1 -> Layer2Switch-1 Gi0/3 лишний trunk VLAN999

Gi0/2 -> Layer2Switch-2 Gi0/2 лишний trunk VLAN999
Gi0/3 -> Layer2Switch-2 Gi0/3 лишний trunk VLAN999
Gi1/0 -> PC1 access VLAN20
Gi1/1 -> PC2 access VLAN333

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname Layer2Switch-3
Layer2Switch-3(config)#
Layer2Switch-3(config)#vlan 999
Layer2Switch-3(config-vlan)# name UNUSED_LINKS
Layer2Switch-3(config-vlan)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet0/0
Layer2Switch-3(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-1_Gi0/2
Layer2Switch-3(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-3(config-if)# switchport mode trunk
Layer2Switch-3(config-if)# switchport trunk native vlan 333
Layer2Switch-3(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet1/0
Layer2Switch-3(config-if)# description PC1_ACCESS_VLAN20
Layer2Switch-3(config-if)# switchport mode access
Layer2Switch-3(config-if)# switchport access vlan 20
Layer2Switch-3(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single
host. Connecting hubs, concentrators, switches, bridges, etc...
to this
interface when portfast is enabled, can cause temporary bridging
loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only
have effect when the interface is in a non-trunking mode.
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet1/1
Layer2Switch-3(config-if)# description PC2_ACCESS_VLAN333
Layer2Switch-3(config-if)# switchport mode access
Layer2Switch-3(config-if)# switchport access vlan 333
Layer2Switch-3(config-if)# spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only have effect when the interface is in a non-trunking mode.

```
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet0/1
Layer2Switch-3(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gi0/3
Layer2Switch-3(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-3(config-if)# switchport mode trunk
Layer2Switch-3(config-if)# switchport trunk native vlan 999
Layer2Switch-3(config-if)# switchport trunk allowed vlan 999
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet0/2
Layer2Switch-3(config-if)# description
UNUSED_LINK_TO_Layer2Switch-2_Gi0/2
Layer2Switch-3(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-3(config-if)# switchport mode trunk
Layer2Switch-3(config-if)# switchport trunk native vlan 999
Layer2Switch-3(config-if)# switchport trunk allowed vlan 999
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#interface GigabitEthernet0/3
Layer2Switch-3(config-if)# description
UNUSED_LINK_TO_Layer2Switch-2_Gi0/3
Layer2Switch-3(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-3(config-if)# switchport mode trunk
Layer2Switch-3(config-if)# switchport trunk native vlan 999
Layer2Switch-3(config-if)# switchport trunk allowed vlan 999
Layer2Switch-3(config-if)# no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#
Layer2Switch-3(config)#end
Layer2Switch-3#write memory
show vlan brief
show interfaces trunk
```

```
Layer2Switch-3#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
20	VLAN20	active	Gi1/0
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	Gi1/1
999	UNUSED_LINKS	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

```
Layer2Switch-3#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	on	802.1q	trunking	999
Gi0/2	on	802.1q	trunking	999
Gi0/3	on	802.1q	trunking	999

Port	Vlans allowed on trunk
Gi0/0	20,333
Gi0/1	999
Gi0/2	999
Gi0/3	999

Port	Vlans allowed and active in management domain
Gi0/0	20,333
Gi0/1	999
Gi0/2	999
Gi0/3	999

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20,333
Gi0/1	999
Gi0/2	none
Gi0/3	none

На свичах, подключенных к компьютерам, дополнительно использовался spanning-tree portfast, чтобы access-порт, к которому подключен компьютер, сразу переходил в состояние forwarding и не ждал STP.

Layer2Switch-4

Gi0/0 -> Layer2Switch-1 Gi1/0 лишний trunk VLAN999

Gi0/1 -> Layer2Switch-1 Gi1/1 лишний trunk VLAN999

Gi0/2 -> Layer2Switch-2 Gi1/0 рабочий trunk

Gi0/3 -> Layer2Switch-2 Gi1/1 лишний trunk VLAN999

Gi1/0 -> PC3 access VLAN20

Gi1/1 -> PC4 access VLAN333

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#

```

Switch(config)#hostname Layer2Switch-4
Layer2Switch-4(config)#
Layer2Switch-4(config)#vlan 999
Layer2Switch-4(config-vlan)# name UNUSED_LINKS
Layer2Switch-4(config-vlan)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet0/2
Layer2Switch-4(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-2_Gil/0
Layer2Switch-4(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)# switchport mode trunk
Layer2Switch-4(config-if)# switchport trunk native vlan 333
Layer2Switch-4(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet1/0
Layer2Switch-4(config-if)# description PC3_ACCESS_VLAN20
Layer2Switch-4(config-if)# switchport mode access
Layer2Switch-4(config-if)# switchport access vlan 20
Layer2Switch-4(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single
host. Connecting hubs, concentrators, switches, bridges, etc...
to this
interface when portfast is enabled, can cause temporary bridging
loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only
have effect when the interface is in a non-trunking mode.
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet1/1
Layer2Switch-4(config-if)# description PC4_ACCESS_VLAN333
Layer2Switch-4(config-if)# switchport mode access
Layer2Switch-4(config-if)# switchport access vlan 333
Layer2Switch-4(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single
host. Connecting hubs, concentrators, switches, bridges, etc...
to this
interface when portfast is enabled, can cause temporary bridging
loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only

```



```
have effect when the interface is in a non-trunking mode.
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet0/0
Layer2Switch-4(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gil/0
Layer2Switch-4(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)# switchport mode trunk
Layer2Switch-4(config-if)# switchport trunk native vlan 999
Layer2Switch-4(config-if)# switchport trunk allowed vlan 999
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet0/1
Layer2Switch-4(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gil/1
Layer2Switch-4(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)# switchport mode trunk
Layer2Switch-4(config-if)# switchport trunk native vlan 999
Layer2Switch-4(config-if)# switchport trunk allowed vlan 999
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#interface GigabitEthernet0/3
Layer2Switch-4(config-if)# description
UNUSED_LINK_TO_Layer2Switch-2_Gil/1
Layer2Switch-4(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-4(config-if)# switchport mode trunk
Layer2Switch-4(config-if)# switchport trunk native vlan 999
Layer2Switch-4(config-if)# switchport trunk allowed vlan 999
Layer2Switch-4(config-if)# no shutdown
Layer2Switch-4(config-if)#exit
Layer2Switch-4(config)#
Layer2Switch-4(config)#end
Layer2Switch-4#write memory
Building configuration...
Compressed configuration from 5927 bytes to 2375 bytes
*May 26 16:20:15.553: %SYS-5-CONFIG_I: Configured from console by
console
```

show vlan brief

show interfaces trunk

```
disk successfully.show vlan brief

VLAN Name                Status    Ports
-----
1    default                active
20   VLAN20                  active    Gi1/0
100  VLAN100                 active
200  VLAN0200                active
300  VLAN0300                active
333  VLAN333                 active    Gi1/1
999  UNUSED_LINKS            active
1002 fddi-default             act/unsup
1003 trcrf-default          act/unsup
1004 fddinet-default        act/unsup
1005 trbrf-default         act/unsup
Layer2Switch-4#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    999
Gi0/1     on        802.1q         trunking    999
Gi0/2     on        802.1q         trunking    333
Gi0/3     on        802.1q         trunking    999

Port      Vlans allowed on trunk
Gi0/0     999
Gi0/1     999
Gi0/2     20,333
Gi0/3     999

Port      Vlans allowed and active in management domain
Gi0/0     999
Gi0/1     999
Gi0/2     20,333
Gi0/3     999

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     999
Gi0/1     999
Gi0/2     20,333
Gi0/3     999
```

Layer2Switch-5

Gi0/0 -> Layer2Switch-1 Gi1/2 лишний trunk VLAN999

Gi0/1 -> Layer2Switch-1 Gi1/3 лишний trunk VLAN999

Gi0/2 -> Layer2Switch-2 Gi1/2 рабочий trunk

Gi0/3 -> Layer2Switch-2 Gi1/3 лишний trunk VLAN999

Gi1/0 -> PC5 access VLAN20

Gi1/1 -> PC6 access VLAN333

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#

Switch(config)#hostname Layer2Switch-5

Layer2Switch-5(config)#

```

Layer2Switch-5(config)#vlan 999
Layer2Switch-5(config-vlan)# name UNUSED_LINKS
Layer2Switch-5(config-vlan)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet0/2
Layer2Switch-5(config-if)# description
WORKING_TRUNK_TO_Layer2Switch-2_Gil/2
Layer2Switch-5(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-5(config-if)# switchport mode trunk
Layer2Switch-5(config-if)# switchport trunk native vlan 333
Layer2Switch-5(config-if)# switchport trunk allowed vlan 20,333
Layer2Switch-5(config-if)# no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet1/0
Layer2Switch-5(config-if)# description PC5_ACCESS_VLAN20
Layer2Switch-5(config-if)# switchport mode access
Layer2Switch-5(config-if)# switchport access vlan 20
Layer2Switch-5(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single
  host. Connecting hubs, concentrators, switches, bridges, etc...
to this
  interface when portfast is enabled, can cause temporary bridging
loops.
  Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only
  have effect when the interface is in a non-trunking mode.
Layer2Switch-5(config-if)# no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet1/1
Layer2Switch-5(config-if)# description PC6_ACCESS_VLAN333
Layer2Switch-5(config-if)# switchport mode access
Layer2Switch-5(config-if)# switchport access vlan 333
Layer2Switch-5(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a
single
  host. Connecting hubs, concentrators, switches, bridges, etc...
to this
  interface when portfast is enabled, can cause temporary bridging
loops.
  Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only
  have effect when the interface is in a non-trunking mode.
Layer2Switch-5(config-if)# no shutdown

```

```
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet0/0
Layer2Switch-5(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gil/2
Layer2Switch-5(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-5(config-if)# switchport mode trunk
Layer2Switch-5(config-if)# switchport trunk native vlan 999
Layer2Switch-5(config-if)# switchport trunk allowed vlan 999
Layer2Switch-5(config-if)# no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet0/1
Layer2Switch-5(config-if)# description
UNUSED_LINK_TO_Layer2Switch-1_Gil/3
Layer2Switch-5(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-5(config-if)# switchport mode trunk
Layer2Switch-5(config-if)# switchport trunk native vlan 999
Layer2Switch-5(config-if)# switchport trunk allowed vlan 999
Layer2Switch-5(config-if)# no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#interface GigabitEthernet0/3
Layer2Switch-5(config-if)# description
UNUSED_LINK_TO_Layer2Switch-2_Gil/3
Layer2Switch-5(config-if)# switchport trunk encapsulation dot1q
Layer2Switch-5(config-if)# switchport mode trunk
Layer2Switch-5(config-if)# switchport trunk native vlan 999
Layer2Switch-5(config-if)# switchport trunk allowed vlan 999
Layer2Switch-5(config-if)# no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#
Layer2Switch-5(config)#end
Layer2Switch-5#write memory
*May 26 16:18:19.449: %SYS-5-CONFIG_I: Configured from console by
console
Building configuration...
```

```
show vlan brief
show interfaces trunk
```

```
disk successfully.show vlan brief

VLAN Name                Status    Ports
-----
1    default                active
20   VLAN20                  active    Gi1/0
100  VLAN100                  active
200  VLAN0200                 active
300  VLAN0300                 active
333  VLAN333                  active    Gi1/1
999  UNUSED_LINKS             active
1002 fddi-default             act/unsup
1003 trcrf-default           act/unsup
1004 fddinet-default         act/unsup
1005 trbrf-default          act/unsup
Layer2Switch-5#show interfaces trunk

Port      Mode          Encapsulation  Status        Native vlan
-----
Gi0/0     on            802.1q         trunking      999
Gi0/1     on            802.1q         trunking      999
Gi0/2     on            802.1q         trunking      333
Gi0/3     on            802.1q         trunking      999

Port      Vlans allowed on trunk
-----
Gi0/0     999
Gi0/1     999
Gi0/2     20,333
Gi0/3     999

Port      Vlans allowed and active in management domain
-----
Gi0/0     999
Gi0/1     999
Gi0/2     20,333
Gi0/3     999

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gi0/0     999
Gi0/1     none
Gi0/2     20,333
Gi0/3     none
Layer2Switch-5#
```

Теперь переходим к заданию айпи адресов у компьютеров:

PC1> ip 192.168.20.1/24

PC2> ip 192.168.33.2/24

PC3> ip 192.168.20.3/24

PC4> ip 192.168.33.4/24

PC5> ip 192.168.20.5/24

PC6> ip 192.168.33.6/24

PC1 PC2

```
PC1> ip 192.168.20.1/24
Checking for duplicate address...
PC1 : 192.168.20.1 255.255.255.0

PC1> wr
Saving startup configuration to startup.vpc
. done

PC2> ip 192.168.33.2/24
Checking for duplicate address...
PC2 : 192.168.33.2 255.255.255.0

PC2> wr
Saving startup configuration to startup.vpc
. done
```

PC3 PC4

<pre>PC3> ip 192.168.20.3/24 Checking for duplicate address... PC3 : 192.168.20.3 255.255.255.0 PC3> wr Saving startup configuration to startup.vpc . done PC3> █</pre>	<pre>PC4> ip 192.168.33.4/24 Checking for duplicate address... PC4 : 192.168.33.4 255.255.255.0 PC4> wr Saving startup configuration to startup.vpc . done PC4> █</pre>
--	--

PC5 PC6

<pre>PC5> ip 192.168.20.5/24 Checking for duplicate address... PC5 : 192.168.20.5 255.255.255.0 PC5> wr Saving startup configuration to startup.vpc . done</pre>	<pre>PC6> ip 192.168.33.6/24 Checking for duplicate address... PC6 : 192.168.33.6 255.255.255.0 PC6> wr Saving startup configuration to startup.vpc . done</pre>
---	---

2. Проверка доступности компьютеров:

VLAN20:

PC1> ping 192.168.20.3

PC1> ping 192.168.20.5

PC3> ping 192.168.20.1

PC3> ping 192.168.20.5

PC5> ping 192.168.20.1

PC5> ping 192.168.20.3

PC1

```
PC1> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=15.158 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=83.719 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=31.673 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=18.144 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=34.656 ms

PC1> ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=28.886 ms
84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=39.779 ms
84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=42.850 ms
84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=73.095 ms
84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=53.979 ms
```

PC3

```
PC3> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=42.660 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=38.864 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=10.643 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=13.600 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=27.674 ms

PC3> ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=13.122 ms
84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=25.947 ms
84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=16.647 ms
84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=48.535 ms
84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=18.894 ms
```

PC5

```
PC5> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=15.587 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=24.399 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=45.780 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=13.421 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=35.396 ms

PC5> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=34.018 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=14.554 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=38.090 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=29.946 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=38.158 ms
```

Соединение между компьютерами VLAN20 есть.

VLAN333:

PC2> ping 192.168.33.4

PC2> ping 192.168.33.6

PC4> ping 192.168.33.2

PC4> ping 192.168.33.6

PC6> ping 192.168.33.2

PC6> ping 192.168.33.4

PC2

```
PC2> ping 192.168.33.4

84 bytes from 192.168.33.4 icmp_seq=1 ttl=64 time=33.684 ms
84 bytes from 192.168.33.4 icmp_seq=2 ttl=64 time=41.242 ms
84 bytes from 192.168.33.4 icmp_seq=3 ttl=64 time=30.156 ms
84 bytes from 192.168.33.4 icmp_seq=4 ttl=64 time=14.074 ms
84 bytes from 192.168.33.4 icmp_seq=5 ttl=64 time=14.982 ms

PC2> ping 192.168.33.6

84 bytes from 192.168.33.6 icmp_seq=1 ttl=64 time=46.985 ms
84 bytes from 192.168.33.6 icmp_seq=2 ttl=64 time=18.810 ms
84 bytes from 192.168.33.6 icmp_seq=3 ttl=64 time=76.859 ms
84 bytes from 192.168.33.6 icmp_seq=4 ttl=64 time=21.588 ms
84 bytes from 192.168.33.6 icmp_seq=5 ttl=64 time=60.508 ms
```

PC4

```
PC4> ping 192.168.33.2

84 bytes from 192.168.33.2 icmp_seq=1 ttl=64 time=33.303 ms
84 bytes from 192.168.33.2 icmp_seq=2 ttl=64 time=42.764 ms
84 bytes from 192.168.33.2 icmp_seq=3 ttl=64 time=43.701 ms
84 bytes from 192.168.33.2 icmp_seq=4 ttl=64 time=35.311 ms
84 bytes from 192.168.33.2 icmp_seq=5 ttl=64 time=28.768 ms

PC4> ping 192.168.33.6

84 bytes from 192.168.33.6 icmp_seq=1 ttl=64 time=58.457 ms
84 bytes from 192.168.33.6 icmp_seq=2 ttl=64 time=37.912 ms
84 bytes from 192.168.33.6 icmp_seq=3 ttl=64 time=10.337 ms
84 bytes from 192.168.33.6 icmp_seq=4 ttl=64 time=62.233 ms
84 bytes from 192.168.33.6 icmp_seq=5 ttl=64 time=26.722 ms
```

PC6

```
PC6> ping 192.168.33.2

84 bytes from 192.168.33.2 icmp_seq=1 ttl=64 time=39.530 ms
84 bytes from 192.168.33.2 icmp_seq=2 ttl=64 time=49.476 ms
84 bytes from 192.168.33.2 icmp_seq=3 ttl=64 time=17.439 ms
84 bytes from 192.168.33.2 icmp_seq=4 ttl=64 time=58.921 ms
84 bytes from 192.168.33.2 icmp_seq=5 ttl=64 time=15.587 ms

PC6> ping 192.168.33.4

84 bytes from 192.168.33.4 icmp_seq=1 ttl=64 time=16.069 ms
84 bytes from 192.168.33.4 icmp_seq=2 ttl=64 time=28.468 ms
84 bytes from 192.168.33.4 icmp_seq=3 ttl=64 time=14.773 ms
84 bytes from 192.168.33.4 icmp_seq=4 ttl=64 time=36.798 ms
84 bytes from 192.168.33.4 icmp_seq=5 ttl=64 time=33.766 ms
```

Соединение между компьютерами VLAN333 есть

Проверка соединения между разными VLAN

PC1> ping 192.168.33.2
PC1> ping 192.168.33.4
PC1> ping 192.168.33.6

PC2> ping 192.168.20.1
PC2> ping 192.168.20.3
PC2> ping 192.168.20.5

PC1

```
PC1> ping 192.168.33.2
No gateway found

PC1> ping 192.168.33.4
No gateway found

PC1> ping 192.168.33.6
No gateway found
```

PC2

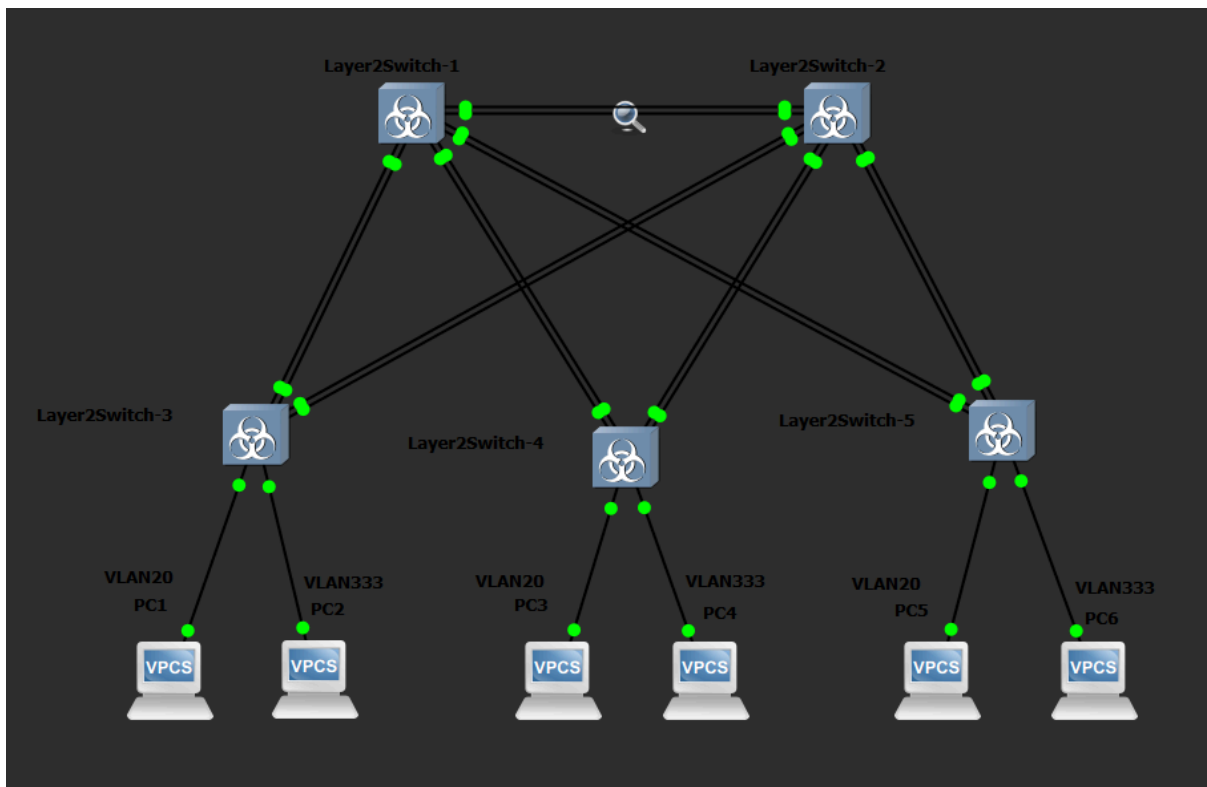
```
PC2> ping 192.168.20.1
No gateway found

PC2> ping 192.168.20.3
No gateway found

PC2> ping 192.168.20.5
No gateway found
```

Соединения нет.

3. Запустил захват пакетов на trunk линке
Layer2Switch-1 Ethernet0 ↔ Layer2Switch-2 Ethernet0



Пропинговал PC5 с PC1 чтобы пакеты прошли

PC1 -> SW3 -> SW1 -> SW2 -> SW5 -> PC5

ping 192.168.20.5

```
PC1> ping 192.168.20.5

84 bytes from 192.168.20.5: icmp_seq=1 ttl=64 time=33.406 ms
84 bytes from 192.168.20.5: icmp_seq=2 ttl=64 time=44.049 ms
84 bytes from 192.168.20.5: icmp_seq=3 ttl=64 time=25.277 ms
84 bytes from 192.168.20.5: icmp_seq=4 ttl=64 time=13.231 ms
84 bytes from 192.168.20.5: icmp_seq=5 ttl=64 time=33.443 ms
```

*Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

vlan

No.	Time	Source	Destination	Protocol	Length	Info
28	30.599738	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
29	30.633271	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
31	33.169369	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
32	33.199178	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
33	34.389725	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.5? Tell 192.168.20.1
34	34.402930	Private_66:68:04	Private_66:68:00	ARP	68	192.168.20.5 is at 00:50:79:66:68:04
35	34.422770	192.168.20.1	192.168.20.5	ICMP	102	Echo (ping) request id=0x6bce, seq=1/256, ttl=64 (reply in 36)
36	34.440040	192.168.20.5	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6bce, seq=1/256, ttl=64 (request in 35)
37	35.452870	192.168.20.1	192.168.20.5	ICMP	102	Echo (ping) request id=0x6cce, seq=2/512, ttl=64 (reply in 38)
38	35.466471	192.168.20.5	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6cce, seq=2/512, ttl=64 (request in 37)
40	36.164258	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
41	36.228034	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
42	36.492304	192.168.20.1	192.168.20.5	ICMP	102	Echo (ping) request id=0x6dce, seq=3/768, ttl=64 (reply in 43)
43	36.514650	192.168.20.5	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6dce, seq=3/768, ttl=64 (request in 42)
44	37.521236	192.168.20.1	192.168.20.5	ICMP	102	Echo (ping) request id=0x6ece, seq=4/1024, ttl=64 (reply in 45)
45	37.529047	192.168.20.5	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6ece, seq=4/1024, ttl=64 (request in 44)
46	38.541225	192.168.20.1	192.168.20.5	ICMP	102	Echo (ping) request id=0x6fce, seq=5/1280, ttl=64 (reply in 47)
47	38.561807	192.168.20.5	192.168.20.1	ICMP	102	Echo (ping) reply id=0x6fce, seq=5/1280, ttl=64 (request in 46)
48	39.237312	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
49	39.282267	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
50	41.512206	0c:5a:b8:a5:00:00	CDP/VTP/DTP/PAgP/UD	DTP	62	Dynamic Trunk Protocol

Frame 33: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

- Destination: Broadcast (ff:ff:ff:ff:ff:ff)
- Source: Private_66:68:00 (00:50:79:66:68:00)
- Type: 802.1Q Virtual LAN (0x8100)
- [Stream index: 5]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20

Address Resolution Protocol (request)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: request (1)
- Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
- Sender IP address: 192.168.20.1
- Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
- Target IP address: 192.168.20.5

Видим поле 802.1Q Virtual LAN с ID:20

Теперь пропингуем через другую сеть пакеты без тега:

Пингую PC6 с PC2:

PC2 -> SW3 -> SW1 -> SW2 -> SW5 -> PC6

ping 192.168.33.6

```
PC2> ping 192.168.33.6

84 bytes from 192.168.33.6 icmp_seq=1 ttl=64 time=32.832 ms
84 bytes from 192.168.33.6 icmp_seq=2 ttl=64 time=41.991 ms
84 bytes from 192.168.33.6 icmp_seq=3 ttl=64 time=36.774 ms
84 bytes from 192.168.33.6 icmp_seq=4 ttl=64 time=17.022 ms
84 bytes from 192.168.33.6 icmp_seq=5 ttl=64 time=45.063 ms
```

Линк тот же

*Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length Info	
1	0.000000	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
2	0.001419	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
3	2.476040	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
4	2.505118	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
5	3.697704	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.33.6? Tell 192.168.33.2
6	3.758806	Private_66:68:05	Private_66:68:01	ARP	64	192.168.33.6 is at 00:50:79:66:68:05
7	3.767651	192.168.33.2	192.168.33.6	ICMP	98	Echo (ping) request id=0x34d0, seq=1/256, ttl=64 (reply in 8)
8	3.784435	192.168.33.6	192.168.33.2	ICMP	98	Echo (ping) reply id=0x34d0, seq=1/256, ttl=64 (request in 7)
9	4.637612	0c:72:4f:52:00:00	0c:72:4f:52:00:00	LOOP	60	Reply
10	4.796923	192.168.33.2	192.168.33.6	ICMP	98	Echo (ping) request id=0x35d0, seq=2/512, ttl=64 (reply in 11)
11	4.828904	192.168.33.6	192.168.33.2	ICMP	98	Echo (ping) reply id=0x35d0, seq=2/512, ttl=64 (request in 10)
12	4.876852	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
13	4.914151	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
14	5.537537	0c:72:4f:52:00:00	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
15	5.843499	192.168.33.2	192.168.33.6	ICMP	98	Echo (ping) request id=0x36d0, seq=3/768, ttl=64 (reply in 16)
16	5.868398	192.168.33.6	192.168.33.2	ICMP	98	Echo (ping) reply id=0x36d0, seq=3/768, ttl=64 (request in 15)
17	5.887789	0c:5a:b8:a5:00:00	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
18	6.878681	192.168.33.2	192.168.33.6	ICMP	98	Echo (ping) request id=0x37d0, seq=4/1024, ttl=64 (reply in 19)
19	6.890980	192.168.33.6	192.168.33.2	ICMP	98	Echo (ping) reply id=0x37d0, seq=4/1024, ttl=64 (request in 18)
20	7.711760	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
21	7.774855	0c:72:4f:52:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:0a:fb:6e:00:00 Cost = 4 Port = 0x8001
22	7.911149	192.168.33.2	192.168.33.6	ICMP	98	Echo (ping) request id=0x38d0, seq=5/1280, ttl=64 (reply in 23)
23	7.928961	192.168.33.6	192.168.33.2	ICMP	98	Echo (ping) reply id=0x38d0, seq=5/1280, ttl=64 (request in 22)
24	8.263628	0c:5a:b8:a5:00:00	CDP/VTP/DTP/PagP/UD...	CDP	432	Device ID: Layer2Switch-1 Port ID: GigabitEthernet0/0

▶ Frame 5: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0	0000	ff ff ff ff ff f
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010	08 00 06 04 00 0
▶ Destination: Broadcast (ff:ff:ff:ff:ff:ff)	0020	ff ff ff ff ff f
▶ Source: Private_66:68:01 (00:50:79:66:68:01)	0030	00 00 00 00 00 0
Type: ARP (0x0806)		
[Stream index: 1]		
Padding: 00000000000000000000000000000000		
Frame check sequence: 0x00000000 [unverified]		
[FCS Status: Unverified]		
▼ Address Resolution Protocol (request)		
Hardware type: Ethernet (1)		
Protocol type: IPv4 (0x0800)		
Hardware size: 6		
Protocol size: 4		
Opcode: request (1)		
Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)		
Sender IP address: 192.168.33.2		
Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)		
Target IP address: 192.168.33.6		

Как мы видим, здесь просто видны поля Ethernet II.

VLAN333 настроена как native VLAN на trunk, который на 802.1Q trunk используется для untagged-трафика

4. Сохранил конфиги командами:

enable

copy running-config startup-config

terminal length 0

show running-config

и для PC

show ip

save